Clustering

Ejercicio 1 - Guía

01.1

$$d(I_1, I_2) = \sqrt{(2,1-2,5)^2 + (1,4-2)^2 + (3-4)^2} = 1,232$$

$$d(I_1, I_3) = \sqrt{(2,1-3,2)^2 + (1,4-0,4)^2 + (3-5)^2} = 2,491$$

$$d(I_2, I_2) = 0$$

$$d(I_2, I_3) = \sqrt{(2,5-3,2)^2 + (2-0,4)^2 + (4-5)^2} = 2,012$$

$$d(I_3, I_3) = 0$$

$$d(I_4, I_2) = \sqrt{(-1,5-2,5)^2 + (-2-2)^2 + (3-4)^2} = 5,744$$

$$d(I_4, I_3) = \sqrt{(-1,5-3,2)^2 + (-2-0,4)^2 + (3-5)^2} = 5,643$$

$$d(I_5, I_2) = \sqrt{(-3,2-2,5)^2 + (-0,3-2)^2 + (1,2-4)^2} = 6,754$$

$$d(I_5, I_3) = \sqrt{(-3,2-3,2)^2 + (-0,3-0,4)^2 + (1,2-5)^2} = 7,475$$

$$d(I_6, I_2) = \sqrt{(1-2,5)^2 + (1-2)^2 + (1-4)^2} = 3,5$$

$$d(I_6, I_3) = \sqrt{(1-3,2)^2 + (1-0,4)^2 + (1-5)^2} = 4,604$$

Nuevos centroides → Promedio de todas las componentes de las instancias que pertenecen a cada clúster

Centroide C1

$$x1 = \frac{2,1+2,5-3,2+1}{4} = 0,6 \qquad x2 = \frac{1,4+2-0,3+1}{4} = 1,025$$

$$x3 = \frac{3+4+1,2+1}{4} = 2,3$$

$$C1 = (0,6;\ 1,025;\ 2,3)$$

Escaneado con CamScanner

Centroide c2

$x1 = \frac{3,2 - 1,5}{2} = 0,85$ $\quad x2 = \frac{0,4 - 2}{2} = -0,8$ $\quad x3 = \frac{5+3}{2} = 4$

$c2 = (0,85; -0,8; 4)$

2° iteración

$d(l_1, c_1) = \sqrt{(2,1-0,6)^2 + (1,4-1,025)^2 + (3-2,3)^2} = 1,697$ (C1)

$d(l_1, c_2) = \sqrt{(2,1-0,85)^2 + (1,4-(-0,8))^2 + (3-4)^2} = 2,720$

$d(l_2, c_1) = \sqrt{(2,5-0,6)^2 + (2-1,025)^2 + (4-2,3)^2} = 2,729$ (C1)

$d(l_2, c_2) = \sqrt{(2,5-0,85)^2 + (2-(-0,8))^2 + (4-2,3)^2} = 3,667$

$d(l_3, c_1) = \sqrt{(3,2-0,6)^2 + (0,4-1,025)^2 + (5-2,3)^2} = 3,800$

$d(l_3, c_2) = \sqrt{(3,2-0,85)^2 + (0,4-(-0,8))^2 + (5-4)^2} = 2,821$ (C2)

$d(l_4, c_1) = \sqrt{(-1,5-0,6)^2 + (-2-1,025)^2 + (3-2,3)^2} = 3,748$

$d(l_4, c_2) = \sqrt{(-1,5-0,85)^2 + (-2-(-0,8))^2 + (3-4)^2} = 2,821$ (C2)

$d(l_5, c_1) = \sqrt{(-3,2-0,6)^2 + (-0,3-1,025)^2 + (1,2-2,3)^2} = 4,172$ (C1)

$d(l_5, c_2) = \sqrt{(-3,2-0,85)^2 + (-0,3-(-0,8))^2 + (1,2-4)^2} = 4,948$

$d(l_6, c_1) = \sqrt{(1-0,6)^2 + (1-1,025)^2 + (1-2,3)^2} = 1,360$ (C1)

$d(l_6, c_2) = \sqrt{(1-0,85)^2 + (1-(-0,8))^2 + (1-4)^2} = 3,501$

Clustering jerárquico

Single Link

- Agrupar instancias tomando la menor distancia

Complete Link

- Agrupar instancias tomando la mayor distancia

Average Link

- Agrupar instancias tomando el promedio de las distancias

Escaneado con CamScanner

	A	B	C	D	E
A	0				
B	9	0			
C	3	7	0		
D	6	5	9	0	
E	11	10	2	8	0

Single - Link

	A	B	C	D	E
A	0				
B	9	0			
C	3	7	0		
D	6	5	9	0	
E	11	10	(2)	8	0

→

	A	B	CE	D
A	0			
B	9	0		
CE	(3)	7	0	
D	6	5	8	0

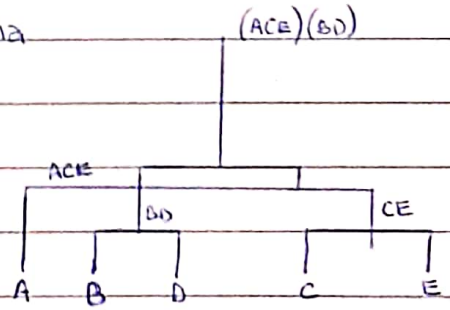
→

	ACE	B	D
ACE	0		
B	7	0	
D	6	(5)	0

→

	ACE	BD
ACE	0	
BD	6	0

Dendograma



Complete Link

	A	B	C	D	E
A	0				
B	9	0			
C	3	7	0		
D	6	5	9	0	
E	11	10	2	8	0

→

	A	B	CE	D
A	0			
B	9	0		
CE	11	10	0	
D	6	5	4	0

→

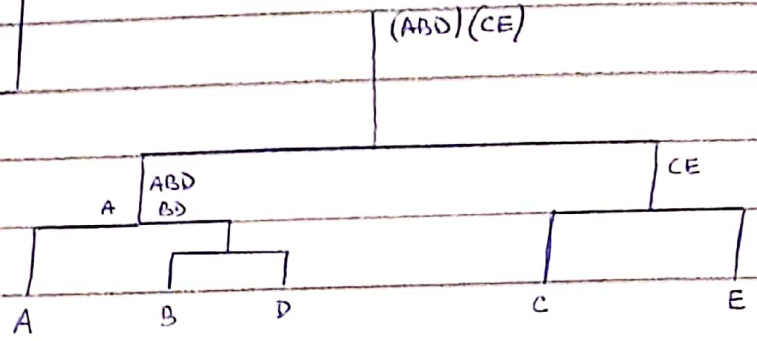
	A	CE	DB
A	0		
CE	11	0	
DB	(9)	10	0

→

Escaneado con CamScanner

	ABD	CE
ABD	0	
CE	11	0

Dendograma



Average Link

	A	B	C	D	E
A	0				
B	9	0			
C	3	7	0		
D	6	5	9	0	
E	11	10	(2)	8	0

→

	A	B	CE	D
A	0			
B	9	0		
CE	7	8,5	0	
D	6	5	8,5	0

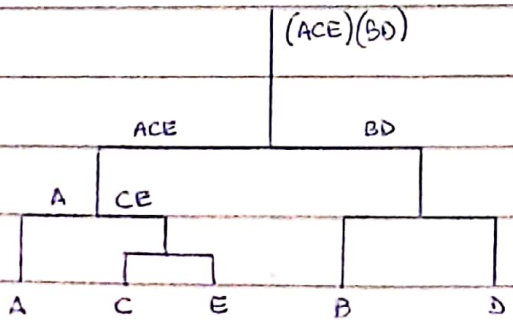
→

	A	BD	CE
A	0		
BD	7,5	0	
CE	(7)	8,5	0

→

	ACE	BD
ACE	0	
BD	8,17	0

Dendograma



Escaneado con CamScanner

Ejercicio Parcial

	A	B	C	D	E
A	0				
B	9.38	0			
C	7.32	17.43	0		
D	16.16	11.22	15.21	0	
E	15.23	12.67	19.88	9.44	0

Single Link

	A	B	C	D	E
A	0				
B	9.38	0			
C	(7.32)	17.43	0		
D	16.16	11.22	15.21	0	
E	15.23	12.67	19.88	9.44	0

→

	AC	B	D	E
AC	0			
B	(9.38)	0		
D	15.21	11.22	0	
E	15.23	12.67	19.88	0

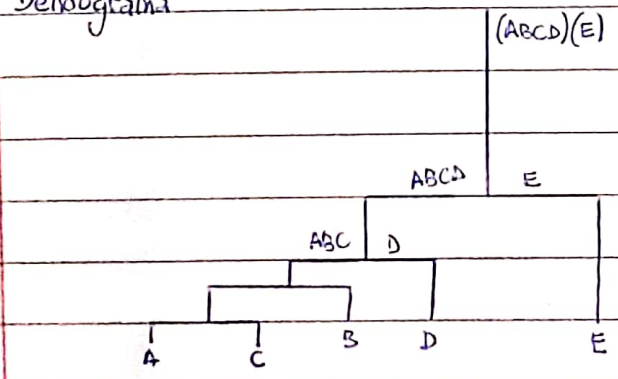
→

	ACB	D	E
ACB	0		
D	(11.22)	0	
E	12.67	19.88	0

→

	ACBD	E
ACBD	0	
E	12.67	0

Dendograma



Escaneado con CamScanner

Complete - Link

	A	B	C	D	E
A	0				
B	9.38	0			
C	7.32	17.43	0		
D	16.16	11.22	15.21	0	
E	15.23	12.67	19.88	9.44	0

→

	A	B	CE	D
A	0			
B	9.38	0		
CE	15.23	17.43	0	
D	16.16	11.22	15.21	0

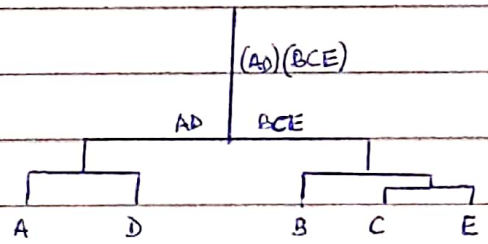
→

	AD	B	CE
AD	0		
B	11.22	0	
CE	15.43	17.43	0

→

	AD	BCE
AD	0	
BCE	15.23	0

Dendograma



Ejercicio final K-means

$I_1 = (1, 1)$ Seeds = I_1, I_2

$I_2 = (3, 4)$

$I_3 = (2, 3)$

$I_4 = (3, 5)$

$I_5 = (4, 2)$

Distancias entre instancias

$d(I_1, I_1) = 0$ (I_1)

$d(I_1, I_2) = \sqrt{13} = 3,60$

$d(I_2, I_1) = \sqrt{13}$

$d(I_2, I_2) = 0$ (I_2)

$d(I_3, I_1) = 2,23$

$d(I_3, I_2) = 1,41$ (I_2)

$d(I_4, I_1) = 4,47$

$d(I_4, I_2) = 1$ (I_2)

$d(I_5, I_1) = 3,16$

$d(I_5, I_2) = 2,23$ (I_2)

Escaneado con CamScanner

Centroides

$C1 = (1, 1)$

$C2 = (3, 3.5)$

$d(I_1, C_1) = 0$ (C1)

$d(I_1, C_2) = 3,20$

$d(I_2, C_1) = 3,60$

$d(I_2, C_2) = 0,5$ (C2)

$d(I_3, C_1) = 2,23$

$d(I_3, C_2) = 1,11$ (C2)

$d(I_4, C_1) = 4,47$

$d(I_4, C_2) = 1,5$ (C2)

$d(I_5, C_1) = 3,16$

$d(I_5, C_2) = 1,80$ (C2)

* No hay reasignación de instancias, por lo que el algoritmo finaliza acá

Escaneado con CamScanner